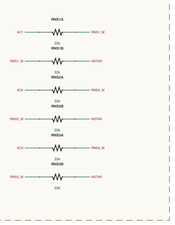
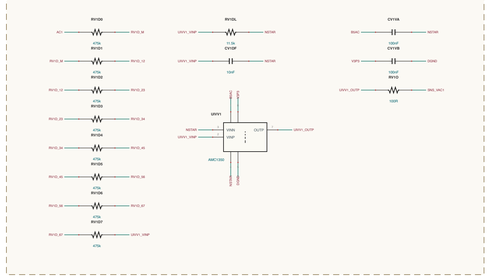
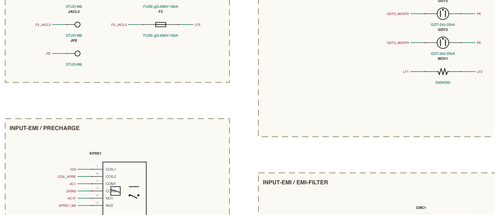
AC-SENSING / STAR



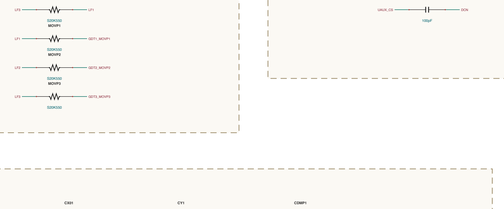
AC-SENSING / SENSE-VAC1



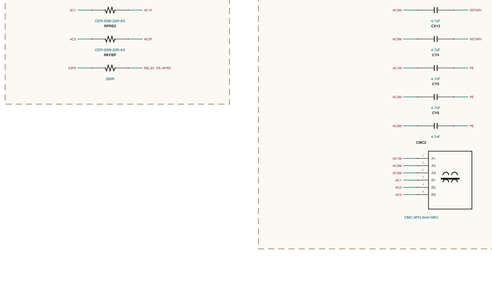
AC-SENSING / SENSE-VAC2



AC-SENSING / SENSE-VAC3

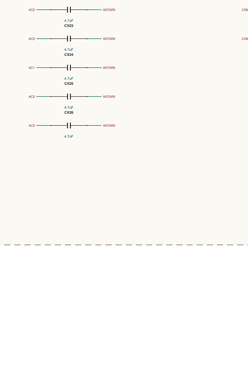


AC-SENSING / SENSE-VBUS



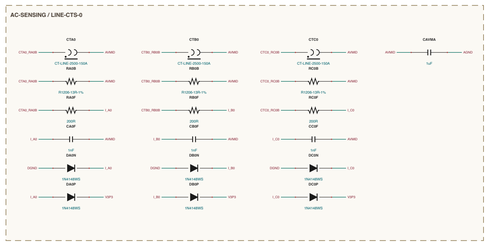
AC-SENSING / SENSE-VMID

AC-SENSING / ISO-BIAS

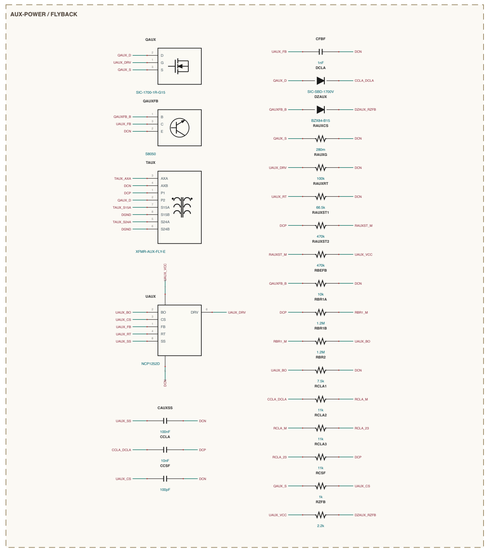


AC-SENSING / NTC

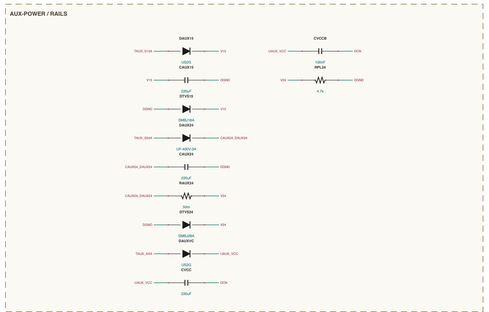
AC-SENSING / LINE-CTS-0



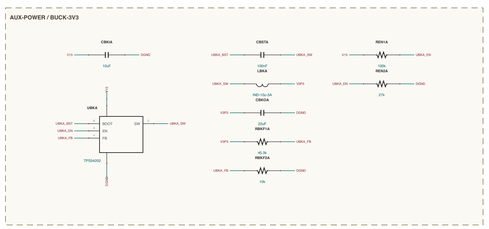
AUX-POWER / FLYBACK



AUX-POWER / RAILS



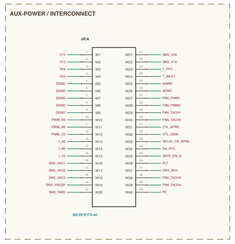
AUX-POWER / BUCK-3V3



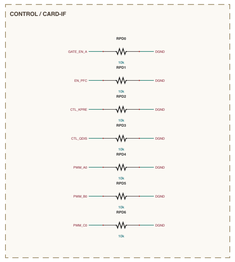
AUX-POWER / FANS



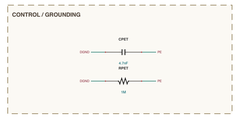
AUX-POWER / INTERCONNECT



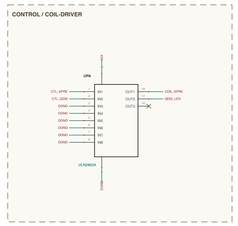
CONTROL / CARD-IF



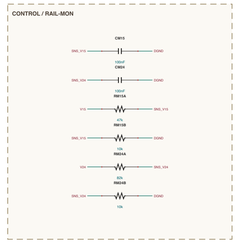
CONTROL / GROUNDING



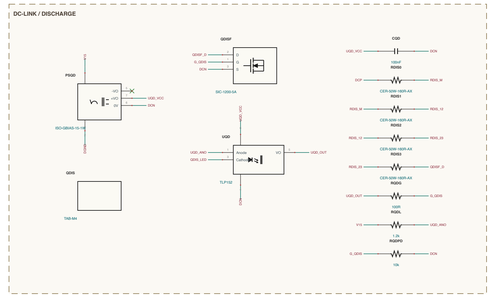
CONTROL / COIL-DRIVER



CONTROL / RAIL-MON



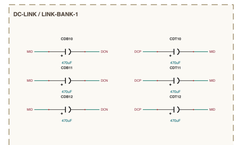
DC-LINK / DISCHARGE



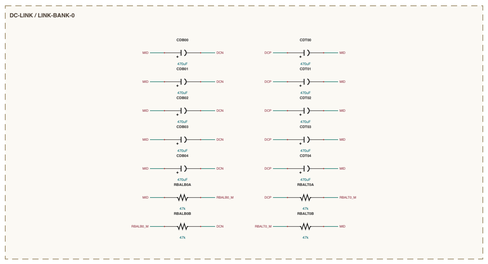
DC-LINK / BUS-STUDS



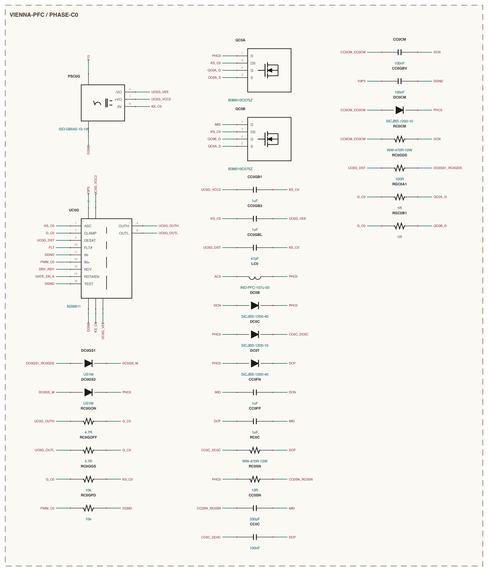
DC-LINK / LINK-BANK-1



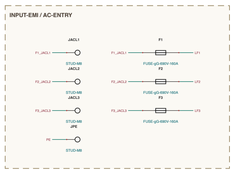
DC-LINK / LINK-BANK-0



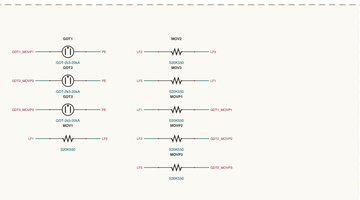
VIENNA-PFC / PHASE-C0



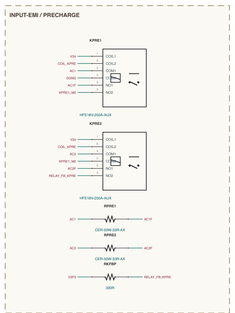
INPUT-EMI / AC-ENTRY



INPUT-EMI / SURGE

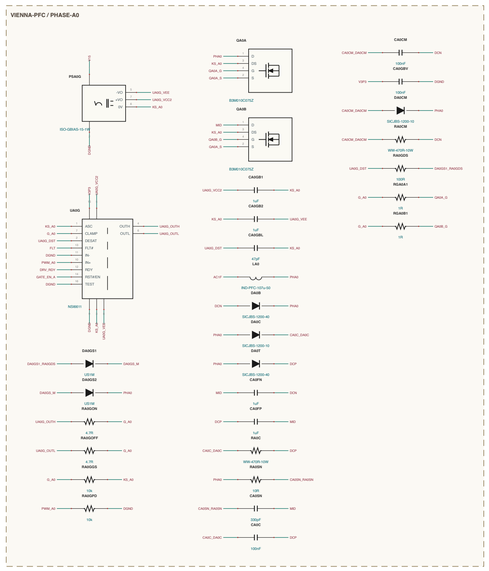


INPUT-EMI / PRECHARGE



INPUT-EMI / EMI-FILTER

VIENNA-PFC / PHASE-A0



SHEET INDEX

50 kW liquid AC-DC (lower) - 1 of 3

AC-SENSING - 9 sections

AUX-POWER - 5 sections

CONTROL - 4 sections

DC-LINK - 4 sections

INPUT-EMI - 4 sections

VIENNA-PFC - 3 sections

rev D.4 - 29 sections - 342 components

NET NAMING

Internal junctions are named for what they join

Uvrw/n_ <PIN> node at that IC pin e.g. UVV2A_VINP

Rxxtem/n_ M midpoint of a series pair e.g. RBALTA_M

Rxxtem/n_ <rm> top between Rxxtem/rm/m e.g. RV1/2_01

all others are explicit design nets

MAGNETICS CONSTRUCTION

Identify per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D1 LA0-LCI: 5x T70 26u aerobust (CATALOG AL37), N=24 +/-1 tol trim -> L3 107uH, ω=65uH @129.5A pk

EMI: CMC1 + 12x X2 4.7uF in 3 star stages + CMC2 + RC damper; no AC-side DM chokes

Cr at TANK sections

DC-Modules 50 kW liquid — AC-DC board (Vienna PFC)

DC-Modules 50 kW liquid - board AC-DC (lower), sheet 1 of 3

2026-09-06 · rev D.4